

Week 8 Worksheet (Solutions)

COMP2048

Semester 1, 2026

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Task 1. *Wolfram Numbering*

- (a) The *majority rule* sets the next state of a cell to 1 if and only if at least two of the three cells in its neighbourhood (left, centre, right) are currently 1. Write out the complete rule table and compute the Wolfram number.
- (b) Reconstruct the rule table for the ECA with Wolfram number 150. Can you express this rule as a single Boolean operation on the three inputs?
- (c) The *complement* of an ECA rule f is the rule $\bar{f}$ defined by $\bar{f}(l, c, r) = 1 - f(1-l, 1-c, 1-r)$. Compute the Wolfram number of the complement of Rule 110.

Solution.

- (a) At least two of three cells are 1 in the neighbourhoods $\{111, 110, 101, 011\}$.

Neighbourhood	111	110	101	100	011	010	001	000
Output	1	1	1	0	1	0	0	0

Wolfram number = $11101000_2 = 128 + 64 + 32 + 8 = \mathbf{232}$.

- (b) $150 = 10010110_2$, so $b_7b_6 \cdots b_0 = 1, 0, 0, 1, 0, 1, 1, 0$:

Neighbourhood	111	110	101	100	011	010	001	000
Output	1	0	0	1	0	1	1	0

The output is 1 exactly when an odd number of the three inputs are 1. This is the **three-input XOR** (parity): $\delta(l, c, r) = l \oplus c \oplus r$.

- (c) To compute $\bar{f}$, we swap $0 \leftrightarrow 1$ in both input and output. The neighbourhood (l, c, r) with index j maps to $(1-l, 1-c, 1-r)$ with index $7-j$. So the complement's output for index j is $1 - b_{7-j}$, where b_i are Rule 110's bits.

Rule 110 = 01101110₂:

Index j	7	6	5	4	3	2	1	0
b_j (Rule 110)	0	1	1	0	1	1	1	0
b_{7-j}	0	1	1	1	0	1	1	0
$1 - b_{7-j}$	1	0	0	0	1	0	0	1

Complement = 10001001₂ = 128 + 8 + 1 = **137**.

Task 2. Evolution on a Finite Ring

Consider Rule 90, whose local rule is $\delta(l, c, r) = l \oplus r$.

- (a) Evolve Rule 90 on a ring of 6 cells from $(0, 1, 0, 0, 0, 0)$ up to $t = 6$.
- (b) At what time step does a previously seen configuration first reappear? What is the period of the resulting cycle?
- (c) Prove that every ECA on a ring of n cells must eventually enter a repeating cycle. What is the maximum number of distinct configurations that can appear before a repeat is guaranteed?

Solution.

- (a) On a ring of 6, cell i 's next value is $c_{(i-1) \bmod 6} \oplus c_{(i+1) \bmod 6}$.

	c_0	c_1	c_2	c_3	c_4	c_5
$t = 0$	0	1	0	0	0	0
$t = 1$	1	0	1	0	0	0
$t = 2$	0	0	0	1	0	1
$t = 3$	1	0	1	0	0	0
$t = 4$	0	0	0	1	0	1
$t = 5$	1	0	1	0	0	0
$t = 6$	0	0	0	1	0	1

- (b) The configuration at $t = 3$ equals the configuration at $t = 1$. The first repeat is at $t = 3$, and the cycle has **period 2** (alternating between $(1, 0, 1, 0, 0, 0)$ and $(0, 0, 0, 1, 0, 1)$). The initial configuration $(0, 1, 0, 0, 0, 0)$ is a transient that does not recur.

- (c) A ring of n binary cells has 2^n possible configurations. The global rule Δ is a deterministic function, so the sequence $c^{(0)}, c^{(1)} = \Delta(c^{(0)}), c^{(2)} = \Delta(c^{(1)}), \dots$ is fully determined by $c^{(0)}$.

By the **pigeonhole principle**, among the first $2^n + 1$ configurations in the sequence, at least two must be identical. Once a configuration repeats, the entire subsequent sequence repeats (since Δ is deterministic), so the system has entered a cycle.

The maximum number of distinct configurations before a repeat is guaranteed is **2^n** .

Task 3. Fixed Points

Find all fixed-point configurations of Rule 30 on a ring of 4 cells, and determine which ECA rules have the all-zeros configuration as a fixed point.

Solution.

(a) Rule 30: $\delta(l, c, r) = l \oplus (c \vee r)$. On a ring of 4, a fixed point (c_0, c_1, c_2, c_3) satisfies:

$$\delta(c_3, c_0, c_1) = c_0$$

$$\delta(c_0, c_1, c_2) = c_1$$

$$\delta(c_1, c_2, c_3) = c_2$$

$$\delta(c_2, c_3, c_0) = c_3$$

With only $2^4 = 16$ configurations, systematic checking is feasible. We verify the three fixed points:

$(0, 0, 0, 0)$: Every neighbourhood is $(0, 0, 0)$, so $\delta = 0 \oplus (0 \vee 0) = 0$. Each cell maps to itself. ✓

$(0, 1, 0, 1)$:

- $\delta(c_3, c_0, c_1) = \delta(1, 0, 1) = 1 \oplus (0 \vee 1) = 0 = c_0$ ✓
- $\delta(c_0, c_1, c_2) = \delta(0, 1, 0) = 0 \oplus (1 \vee 0) = 1 = c_1$ ✓
- $\delta(c_1, c_2, c_3) = \delta(1, 0, 1) = 0 = c_2$ ✓
- $\delta(c_2, c_3, c_0) = \delta(0, 1, 0) = 1 = c_3$ ✓

$(1, 0, 1, 0)$: By symmetry with $(0, 1, 0, 1)$ (both are alternating patterns, and the equations have the same cyclic structure):

- $\delta(0, 1, 0) = 1 = c_0$ ✓
- $\delta(1, 0, 1) = 0 = c_1$ ✓
- $\delta(0, 1, 0) = 1 = c_2$ ✓
- $\delta(1, 0, 1) = 0 = c_3$ ✓

All other configurations fail at least one equation. For example, $(1, 1, 1, 1)$: $\delta(1, 1, 1) = 1 \oplus (1 \vee 1) = 0 \neq 1$.

The complete set of fixed points is: $\{(\mathbf{0}, \mathbf{0}, \mathbf{0}, \mathbf{0}), (\mathbf{0}, \mathbf{1}, \mathbf{0}, \mathbf{1}), (\mathbf{1}, \mathbf{0}, \mathbf{1}, \mathbf{0})\}$.

(b) On the all-zeros configuration, every cell's neighbourhood is $(0, 0, 0)$. The fixed-point condition requires $\delta(0, 0, 0) = 0$, which means $b_0 = 0$.

Since b_0 is the least significant bit of the Wolfram number, the condition is that the **Wolfram number is even**.

This accounts for exactly 128 of the 256 ECA rules.

Task 4. *The Global Map and Gardens of Eden*

Compute Δ for Rule 90 on a ring of 4 cells. Identify Gardens of Eden and find the structural property shared by all image configurations.

Solution.

- (a) Rule 90 on a ring of 4: $\Delta(c)_i = c_{(i-1) \bmod 4} \oplus c_{(i+1) \bmod 4}$.

c	(decimal)	$\Delta(c)$	(decimal)
0000	0	0000	0
0001	1	1010	10
0010	2	0101	5
0011	3	1111	15
0100	4	1010	10
0101	5	0000	0
0110	6	1111	15
0111	7	0101	5
1000	8	0101	5
1001	9	1111	15
1010	10	0000	0
1011	11	1010	10
1100	12	1111	15
1101	13	0101	5
1110	14	1010	10
1111	15	0000	0

- (b) The image of Δ is $\{0000, 0101, 1010, 1111\}$, which has only 4 elements. The remaining **12 configurations are Gardens of Eden**: every configuration not of the form (a, b, a, b) .
- (c) In every output configuration, position 0 equals position 2, and position 1 equals position 3.

This is because $\Delta(c)_0 = c_3 \oplus c_1$ and $\Delta(c)_2 = c_1 \oplus c_3$, which are identical. Similarly, $\Delta(c)_1 = c_0 \oplus c_2 = \Delta(c)_3$.

So the image is restricted to configurations of the form (a, b, a, b) . There are only $2^2 = 4$ such configurations, which is why 12 out of 16 are Gardens of Eden.

Task 5. *Life-like Rules and Neighbourhoods*

- (a) Compute the next generation of the L-tromino under B3/S23 (Moore) and verify the result is a still life.
- (b) What happens to the blinker under B3/S23 with the von Neumann neighbourhood?
- (c) Classify the blinker under B5/S12, B3/S23, and B4/S3.

Solution.

- (a) The L-tromino has live cells at positions $(1, 1)$, $(1, 2)$, $(2, 1)$ in a 4×4 region (row, column from 0).

Moore neighbour counts for each cell:

	col 0	col 1	col 2	col 3
row 0	1	2	2	1
row 1	2	2*	2*	1
row 2	2	2*	3	1
row 3	1	1	1	0

Cells marked * are currently alive.

Under B3/S23: live cells with 2 neighbours survive (S includes 2). Dead cell $(2, 2)$ has exactly 3 live neighbours, so it is born (B includes 3). No other dead cell has 3 neighbours.

The next generation is the 2×2 **block**: live at $(1, 1)$, $(1, 2)$, $(2, 1)$, $(2, 2)$.

Verification that the block is a still life: every live cell in the block has exactly 3 live neighbours $\rightarrow$ survives (S includes 3). The dead cells adjacent to the block have 1 or 2 live neighbours $\rightarrow$ no births (B requires 3). So the block is unchanged. $\checkmark$

- (b) The blinker has live cells at $(1, 1)$, $(1, 2)$, $(1, 3)$.

Von Neumann neighbour counts (4 adjacent cells only):

- $(1, 1)$ live: vN neighbours at $(0, 1)$, $(1, 0)$, $(1, 2)$, $(2, 1)$. Live: $(1, 2)$. Count = 1.
- $(1, 2)$ live: vN neighbours at $(0, 2)$, $(1, 1)$, $(1, 3)$, $(2, 2)$. Live: $(1, 1)$, $(1, 3)$. Count = 2.
- $(1, 3)$ live: vN neighbours at $(0, 3)$, $(1, 2)$, $(1, 4)$, $(2, 3)$. Live: $(1, 2)$. Count = 1.

Under B3/S23: $S = \{2, 3\}$. The end cells have 1 neighbour (not in S), so they **die**. The middle cell has 2 neighbours (in S), so it survives.

Dead cells: the maximum vN count for any dead cell is 2 (for cells $(0, 2)$ and $(2, 2)$), which is not in $B = \{3\}$. No births occur.

Result: the blinker collapses to a single live cell at $(1, 2)$, which then dies at $t = 2$ (it has 0 vN neighbours). The blinker **dies out** under von Neumann.

Why the neighbourhood matters: under Moore, the end cells each have 2 neighbours (the middle cell plus one diagonal cell from the opposite end), which is in S , so they survive. Under von Neumann, diagonal neighbours are excluded, reducing each end cell's count to 1, which is not in S .

(c) Moore neighbour counts for the blinker (live cells at $(1, 1)$, $(1, 2)$, $(1, 3)$):

Live cells: $(1, 1)$: Moore neighbours include $(1, 2)$ as the only live cell. Count = 1. $(1, 2)$: live neighbours $(1, 1)$ and $(1, 3)$. Count = 2. $(1, 3)$: by symmetry with $(1, 1)$. Count = 1.

Dead cells (non-zero counts only):

Count 1: $(0, 0)$, $(0, 4)$, $(1, 0)$, $(1, 4)$, $(2, 0)$, $(2, 4)$
 Count 2: $(0, 1)$, $(0, 3)$, $(2, 1)$, $(2, 3)$
 Count 3: $(0, 2)$, $(2, 2)$

The maximum dead-cell count is 3 (at positions directly above and below the middle cell).

- (i) **B5/S12.** $S = \{1, 2\}$: end cells (count 1) survive, middle (count 2) survives. $B = \{5\}$: no dead cell has 5 neighbours. No births. $\Rightarrow$ **Still life.**
- (ii) **B3/S23.** $S = \{2, 3\}$: end cells (count 1) **die**. Middle (count 2) survives. $B = \{3\}$: dead cells $(0, 2)$ and $(2, 2)$ (count 3) come alive. The pattern becomes a vertical bar at column 2, rows 0–2. This is the blinker rotated 90° , which will rotate back on the next step. $\Rightarrow$ **Oscillator** (period 2).
- (iii) **B4/S3.** $S = \{3\}$: end cells (count 1) die, middle (count 2) dies. $B = \{4\}$: no dead cell has 4 neighbours. $\Rightarrow$ **Dies out** in one step.

Task 6. From 1D to 2D

Write the formal 4-tuple for the Game of Life, count the number of possible 2-state Moore-neighbourhood CAs, and explain why exhaustive classification is infeasible.

Solution.

- (a)
 - $L = \mathbb{Z}^2$ (the infinite integer lattice).
 - $\Sigma = \{0, 1\}$ (dead, alive).
 - $N = ((-1, -1), (-1, 0), (-1, 1), (0, -1), (0, 0), (0, 1), (1, -1), (1, 0), (1, 1))$. This is the Moore neighbourhood *plus the cell itself*: $|N| = 9$.
 - $\delta: \{0, 1\}^9 \rightarrow \{0, 1\}$, defined by: let s be the sum of the 8 neighbours (excluding the centre value c). Then $\delta(\dots) = 1$ iff $(c = 0 \text{ and } s = 3)$ or $(c = 1 \text{ and } s \in \{2, 3\})$.

- (b) The neighbourhood N has 9 entries (8 neighbours plus the cell itself), and each entry is in $\Sigma = \{0, 1\}$. So the number of distinct input patterns to δ is $|\Sigma|^{|N|} = 2^9 = 512$.

A rule is a function that assigns an output (0 or 1) to each of these 512 input patterns. There are 2 choices per pattern, so the total number of distinct 2-state Moore-neighbourhood CAs is $2^{512} \approx 1.3 \times 10^{154}$.

Compare with ECAs: an ECA neighbourhood has $|N| = 3$, giving $2^3 = 8$ input patterns and $2^8 = 256$ rules. The jump from 8 input patterns to 512 makes the rule space explode from 2^8 to 2^{512} .

- (c) Wolfram classified all 256 ECA rules by running each one on many initial conditions and observing the long-term behaviour. This was feasible because 256 is a small number.

For 2D Moore-neighbourhood CAs, there are 2^{512} possible rules. To put this in perspective: $2^{512} \approx 10^{154}$, which exceeds the number of atoms in the observable universe ($\approx 10^{80}$) by roughly 74 orders of magnitude. You cannot enumerate them, let alone simulate and classify each one.

Even restricting to Life-like CAs (where the rule depends only on the neighbour *count* and current state, as in B/S notation) gives $2^9 \times 2^9 = 2^{18} = 262,144$ rules per neighbourhood type. This is tractable in principle but still vastly larger than 256.

Task 7. *Decidability and Cellular Automata*

Determine whether CELL_t is decidable, and discuss whether a program could classify arbitrary CAs into Wolfram classes.

Solution.

- (a) **Yes**, CELL_t is Turing-decidable.

Decider D : “On input $\langle R, c, i, t \rangle$:

1. Simulate ECA rule R on configuration c for exactly t steps.
2. Check whether cell i is in state 1 at time t .
3. If yes, ACCEPT. Otherwise, REJECT.”

Each simulation step updates every cell in constant time per cell, and there are exactly t steps. The simulation always terminates, so D is a decider.

This parallels HALT_{2048} from the Week 7 worksheet: both are decidable because a **bounded simulation** always terminates.

- (b) **No**, such a program cannot exist for arbitrary CAs.

The fundamental difficulty is distinguishing **Class III (chaotic)** from **Class IV (complex)**. A Class IV rule supports persistent interacting structures (“particles”) that can implement computation. To determine whether a CA is Class IV, you would need to determine whether it supports universal computation, which requires answering a question equivalent to the halting problem.

More precisely: if you could identify Class IV rules, you could identify Turing-complete CAs, and hence determine which CAs have undecidable reachability problems. But this is itself undecidable (by Rice’s theorem applied to the class of Turing-complete systems).

Classes I and II are easier to detect (fixed points and short cycles appear quickly), and Class III can often be identified by statistical properties. The boundary between III and IV is where decidability breaks down.